

RETAKE EXAM

This is a 2 hours exam.

I – A MATCHING MODEL (60% of the points)

We consider the following simple matching model. The cost of posting a vacancy is c . One people are matched, the job delivers a net present value to the firm equal to J , which we treat as exogenous. There is an exogenous probability per unit of time of a job being destroyed equal to s .

In what follows "stable" has a natural economic meaning: an equilibrium situation is stable if, when the choice variable of interest (vacancies) slightly deviates from this equilibrium, the optimal reaction of economic agents tends to bring this variable back to the equilibrium value.

Part A. In this part, the matching function is $m(u, v) = uv$.

1 – What is the equation for the Beveridge curve?

2 – What are the probability of finding a job per unit of time p , and the probability of filling a vacancy q , as functions of u and v ? Can they be written as functions of $\theta = u/v$ only? Why?

3 – Let $u^* = c/J$. Show that

- (i) If $u < u^*$ then $v = 0$ and unemployment gradually rises until it becomes equal to u^* .
- (ii) If $u > u^*$ then unemployment immediately falls to u^* .

What is the vacancy level at the instant when this jump takes place?

4 – (difficult) Does the assumption that J is exogenous make sense in an open economy? In a closed economy? What happens to the trade balance in an open economy if initially $u > u^*$? How would the adjustment take place in an closed economy?

Part B. In this part, the matching function is $m(u, v) = u^2v^2$

5 – Show that at any date t , there are two stable equilibrium outcomes:

- Either unemployment is eliminated instantaneously
- Or there are no vacancies and unemployment gradually increases

6 – In such a case, are we confident that if unemployment gradually increases, this process will continue until everybody is unemployed?

Part C. In this part the matching function is $m(u, v) = \max(\sqrt{uv} - k, 0)$, $k > 0$.

7 – Show that at any date t there are only two stable equilibrium outcomes:

- Either vacancies are equal to zero and unemployment gradually increases (regime I)
- Or they reach a positive number (regime II)

Show that the economy is in regime I if $u_t < 4kc/J$, and can be in either regime if $u_t > 4kc/J$.

8 – How do the vacancy rate v , and labor market tightness θ , vary with u in regime II?

II – TRUE, FALSE, UNCERTAIN (40% of the points)

1 – The business cycle corresponds to the stationary component of economic time series.

- False, there is not such a thing as the stationary component

- Let us consider a times series x_t that we want to decompose into a trend x_t^T and a cycle x_t^C .

- The business cycle is vaguely defined as the periodic but irregular up-and-down movements in economic activity, measured by fluctuations in real GDP and other macroeconomic variables.

- Statistical theory tells us that there is a infinity of decomposition of x_t into a stationary (typically “the cycle”) and a non-stationary (“the trend”).
- Expert analysis (NBER, business analysts, etc...) consider that there is a 2 to 8 years cycle in economic activity.
- One can then somewhat arbitrarily decide that the business cycle will be measured as the sum of all the periodic components of period less than 32 quarters (Hodrick-Prescott filter), or that components between 8 and 32 quarters only will be kept (band pass filter).
- There are several other ways of extracting a business cycles, some more structural (output gap measures), some not (growth rates, linear detrending).
- Generically, the business cycle is one filtered version of x_t .

2 – Given the way it is measured in the data, durable goods consumption is a smooth variable in the business cycle.

- False.
- In the data, what is measured are expenditures. Expenditures on non-durable goods are quite very similar to the *consumption* of non-durable goods (because one has to consume the goods once bought). Therefore, if consumption of non durable goods is smooth, expenditures on non durable goods will also be smooth.
- Things are quite different for durable goods (say cars). Although the consumption of transport services is smooth across time, expenditures on cars are very lumpy.

3 – Calibration of an RBC model is not so far from estimation.

- Uncertain.
- On the one side, calibration is close to estimation. Parameters are chosen so that the model fits some long run facts (share of labor income, capital-output ratio, level of hours worked per period). Often, there are as many long run facts to fit than parameters to set, so that the “ R^2 ” of the model is one as far as those long-run facts are concerned.
- On the other side, some parameters are just set according to some micro econometric studies, without relation to business cycle statistics. Furthermore, no formal evaluation of fit nor test of the model is generally proposed for many calibrated macro models.
- Note that the current practice in DSGE (dynamic stochastic general equilibrium) models is to estimate some (if not all) parameters, using maximum likelihood methods, bayesian methods or minimum distance ones.

4 – In dynamic general equilibrium model, a balanced growth path exists when income effects on labor supply are not too strong.

- Uncertain (imprecise).
- Along a balanced growth path, labor productivity grows at a constant rate. So does the real wage.
- In order for worked hours to stay constant along a BGP, income and substitution effects must exactly offset: income effect implies a decrease in worked hours (when leisure is a normal good, which is always assumed), while substitution effect makes them supplying more labor.
- The two effects exactly offset each other with CRRA preferences of the type

$$U(C_t, \ell_t) = \begin{cases} \frac{C_t^{1-\sigma}}{1-\sigma} v(\ell_t) & \text{for } \sigma \in \mathbb{R}^+ \setminus \{1\} \\ \text{or} \\ \log(C_t) + v(\ell_t) \end{cases}$$